

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 28 September 2026

Team ID: SWTID-2026-1853

Project Name: Auto Ticket Classification using Flow Designer

Maximum Marks: 4 Marks

Technical Architecture:

The deliverable includes the architecture and the information in Table-1 and Table-2 for the proposed Auto Ticket Classification using Flow Designer solution.

Architecture flow (logical view):

Requester / Employee → Service Portal or Ticket Form → Ticket Record (Incident / Case) → Flow Designer Trigger → Classification Rules / Decision Logic → Set Category, Subcategory and Priority → Assignment Group Routing → Agent Work Queue → Status Update and Notifications

Data and platform connections: Ticket and audit data are stored in the service platform. Flow Designer uses configured platform actions and, if enabled, approved integrations or classification services. External interfaces are optional and depend on the deployed instance.

Guidelines:

- Include all processes as application logic / technology blocks: intake, trigger, classification, routing, review, and notification.
- Show infrastructure demarcation: ServiceNow cloud instance or the organization's approved hosted environment; identify any local systems connected through integration.
- Indicate external interfaces such as email, identity services, or approved REST APIs only when configured.
- Indicate ticket records, configuration data, and execution/audit logs as storage components.
- Indicate any machine-learning classification interface only if the project enables and configures such a service.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Requester submits a ticket and agents review or update it.	ServiceNow Service Portal / Employee Center; ServiceNow forms
2.	Application Logic-1	Accept ticket details and create the ticket record.	ServiceNow Incident / Case form and table
3.	Application Logic-2	Start the workflow when a ticket is created or updated.	Flow Designer trigger
4.	Application Logic-3	Evaluate ticket text and fields, then determine category and subcategory.	Flow Designer conditions, decision logic, configured rules
5.	Database	Store ticket details, classification values, assignment, and status.	ServiceNow platform tables (Incident / Case)
6.	Cloud Database	Platform-managed persistence for application records and configuration.	ServiceNow cloud platform data store

7.	File Storage	Optional attachments submitted with tickets.	ServiceNow attachment storage
8.	External API-1	Optional integration to an approved classification or enrichment service.	REST integration / IntegrationHub action, if configured
9.	External API-2	Optional notifications or identity lookup integration.	Email notification / approved identity API, if configured
10.	Machine Learning Model	Optional text classification when an ML capability is enabled.	Configured ServiceNow AI / classification capability; otherwise rule-based Flow Designer logic
11.	Infrastructure (Server / Cloud)	Run workflow and ticketing application.	ServiceNow cloud instance; local systems only through approved integration

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No separate open-source framework is required for the core workflow; platform-native configuration is used.	Flow Designer; optional scripted components only if approved
2.	Security Implementations	Use role-based access, least privilege, authenticated integrations, and platform audit history. Protect credentials and restrict ticket visibility by role.	ServiceNow roles / ACLs, HTTPS, credential records, audit logs; organizational security controls
3.	Scalable Architecture	Workflow is event-triggered and uses platform services. Capacity depends on the configured instance, workflow complexity, and ticket volume.	ServiceNow platform; Flow Designer
4.	Availability	Availability follows the deployed ServiceNow service and organization's operational configuration. Failure paths should leave tickets available for manual handling.	ServiceNow service availability; exception handling and review queue
5.	Performance	Keep conditions concise, avoid unnecessary repeated updates, and route ambiguous tickets for human review. Validate performance against expected ticket volume.	Flow Designer execution monitoring; platform reporting

References:

<https://www.servicenow.com/products/flow-designer.html>

<https://www.servicenow.com/docs/>

<https://c4model.com/>